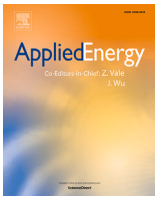
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Operational analysis of the cooling system in a direct liquid-cooled data center: a measurement and simulation study on the impact of supply water temperature

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HIGHLIGHTS

- Analyzed 21 months of operational data from a direct liquid-cooled data center.
- Identified interactions between liquid cooling, air cooling, and IT equipment.
- Developed cross-system model for direct liquid-cooled data centers.
- Evaluated impact of supply water temperature on data center power consumption.
- Identified significant energy savings via dynamic supply temperature control.

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ABSTRACT

To further improve the energy efficiency of direct liquid-cooled data centers, advanced cooling control strategies capable of dynamic adjustments according to varying operational and environmental conditions are required. A critical parameter for optimization in the liquid cooling system is the supply water temperature. This study investigates the impact of the supply water temperature on data center energy performance by analyzing operational data from the direct liquid-cooled HAWK supercomputer, which was operated under different supply temperature settings. The analysis demonstrates that increasing the supply temperature from 17 to 25 °C improves the liquid cooling system's energy efficiency, achieving a power consumption reduction of 63.3 % at an outdoor wet-bulb temperature of 19 °C. However, this also increased heat transfer to the server room air, with the proportion nearly doubling from approximately 2.6 % to 5.0 %, and increased IT equipment power consumption by about 3.15 %. Based on these insights, a data center-level power and thermal model was developed, incorporating the liquid cooling system, air cooling system, IT equipment, and the interactions among these systems. Simulation studies using this model further analyze the impact of supply water temperature on the energy performance of each system and the overall data center. The results highlight the considerable energy-saving potential of dynamic control strategies that adjust the supply water temperature according to outdoor conditions. The findings also highlight the importance of holistic data center modeling for developing effective cooling control strategies and enabling efficient heat reuse integration.

1. Introduction

Driven by the rapid growth of cloud computing, AI workloads, and digital services, the need for energy-efficient data centers is becoming increasingly urgent. Liquid cooling has emerged as an efficient solution for managing the cooling load of high power density servers, owing to the significantly greater heat capacity of liquids compared to air [1,2]. It is anticipated that once the power density in standard 19-inch racks approaches 1000 W per unit, the primary cooling method or even the sole method will be liquid cooling [3]. These include immersion cooling, where the entire server is submerged in a dielectric liquid so that all components are in direct contact with the heat transfer fluid, as well

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Nomenclature

Abbreviations

DC	district cooling
HR	heat reuse
PUE	power usage effectiveness
R^2	R-squared
AHU	air handling unit
BMS	building management system
CDU	cooling distribution unit
CPU	central processing unit
CRAH	computer room air handler
CTS	cooling tower system
CVRMSE	coefficient of variation of the root mean square error
DIMM	dual in-line memory module
EER	energy efficiency ratio
FWS	facilities water system
HLRS	High-Performance Computing Center Stuttgart
IEER	integrated energy efficiency ratio
IT	information technology
LCM	liquid cooling module
MBE	mean bias error
NMBE	normalized mean bias error
PID	proportional-integral-derivative
RMSE	root mean square error
TCS	technology cooling system
UPS	uninterruptible power supply
VR	voltage regulator

Greek symbols

α	constant
β	model parameter
Δp	pressure increase across the pump, Pa
η	pump efficiency
ν	kinematic viscosity, $m^2\ s^{-1}$
ϕ_{aw}	air-to-water mass flow rate ratio
ρ	density, $kg\ m^{-3}$

Latin symbols

$\dot{m}$	mass flow rate, $kg\ s^{-1}$
$\dot{Q}$	heat transfer rate, W
$\dot{V}$	volumetric flow rate, $m^3\ s^{-1}$
A	effective heat exchange area, m^2
c_e	effective switching capacitance, F
c_p	specific heat capacity, $J\ kg^{-1}\ K^{-1}$
dT	temperature change, K
E_{dd}	CPU supply voltage, V
f	frequency, Hz
h	convective heat transfer coefficient, $W\ K^{-1}\ m^{-2}$
k	thermal conductivity of the fluid, $W\ K^{-1}\ m^{-1}$
L	characteristic length, m
N	number of data points
n_{fan}	fan speed, RPM
Nu	Nusselt number
P	power consumption, W
Pr	Prandtl number
R	thermal resistance, $K\ W^{-1}$
Re	Reynolds number
T	temperature, °C
t	time, s
u	fluid velocity, $m\ s^{-1}$
u_{CPU}	CPU utilization rate
UA	overall heat transfer coefficient of the heat exchanger, $W\ K^{-1}$
V	volume, m^3
WBT	wet-bulb temperature, °C

Subscripts

a	air
ACS	air cooling system
c	cold fluid side of the heat exchanger
$cond$	conductive
CP	cold plate
dis	water distribution system
dry	dry cooler system
h	hot fluid side of the heat exchanger
in	inlet conditions
LCS	liquid cooling system
out	outlet conditions
tot	total
w	water in liquid cooling system

as direct liquid cooling, where heat is conducted from the chip to an attached heat transfer component such as a cold plate [2,4].

To fully realize the energy-saving potential of liquid cooling systems, advanced control strategies capable of dynamic adjustments and real-time optimization in response to varying operational and environmental conditions are required. Typically, the primary parameters optimized in a liquid cooling system are the supply water temperature and flow rates [5–10]. To develop effective optimization strategies, it is essential to accurately model the impact of these parameters not only on the energy performance of the liquid cooling system itself but also on other interacting systems.

For instance, increasing the water temperature level within the liquid cooling system can improve the energy efficiency of cooling towers and chillers. According to the ASHRAE Thermal Guidelines for Liquid Cooling, data centers operating at maximum facility supply temperatures above 32 °C may eliminate the need for chillers in most geographical locations [11]. Although higher cooling temperatures seem intuitively advantageous in terms of energy savings and reduced operating costs, such a strategy has several drawbacks.

First, elevated temperatures within the liquid cooling system may negatively affect server performance, since many information technology (IT) systems incorporate built-in protective mechanisms, such as central processing unit (CPU) throttling, to avoid thermal damage when approaching critical temperature thresholds [12]. Second, higher cooling temperatures directly impact the electrical energy consumption of IT equipment, primarily due to increased leakage currents in silicon devices [13]. These leakage currents exhibit significant temperature sensitivity, reported averaged data suggests that a 10 K increase in coolant inlet temperature typically results in a 2.0–4.3 % increase in IT power consumption [14–16].

Additionally, higher water temperatures increase the heat transfer from IT equipment into the server room air. For example, a recent study demonstrated that raising coolant temperature by 20 K reduced the fraction of heat dissipated directly into the liquid coolant by approximately 14.6 %, thus increasing the release of heat into the air in the server room [16]. This shift is undesirable, as air cooling systems typically have lower efficiency compared to liquid cooling systems [11].

Moreover, optimizing flow rates in liquid cooling systems introduces additional complexity. While pump power consumption is directly related to flow rates, changes in flow rates also affect the performance of other systems. Specifically, varying flow rates influence the thermal resistance of IT component heat sinks, such as cold plates [17,18], which in turn affects IT equipment temperatures, and consequently the leakage power. Flow rate adjustments also change the temperature difference

between supply and return streams, as well as the heat transfer coefficients in heat exchangers, thereby influencing temperature levels throughout the liquid cooling system and, consequently, total power consumption.

Given these complex interactions, optimizing the liquid cooling system operation requires a comprehensive understanding not only of the liquid cooling system itself, but also of key dynamics such as heat transfer from IT equipment to the server room air, the thermal behavior and energy performance of the air cooling system, and the temperature dependency of IT leakage power.

Several simulation-based studies have investigated control optimization for liquid cooling systems. He et al. [7,8] developed an analytical power and thermal model for a liquid-cooled server test bench, encompassing components from cold plates to cooling towers. In their first study, they optimized water flow rates and supply temperatures under varying cooling loads and constant ambient conditions, achieving energy savings of up to 10 % compared to baseline operation without optimization [7]. In a follow-up study, they extended the analysis to year-round operation using weather data from Tianjin, China, and demonstrated a 21.3 % reduction in the liquid cooling system's energy consumption [8]. Similarly, Qu et al. [10] developed an analytical model for a test bench, also including components from cold plates to cooling towers. Their study focused on optimizing the cooling tower airflow rate and water flow rates across different loops under varying thermal loads and environmental conditions. Using typical load profiles and weather data from Xi'an, China, they achieved annual energy savings of up to 42.7 %. However, the models employed in these studies are limited to the liquid cooling system and do not account for cross-system interactions. As a result, optimization strategies based on these models tend to drive chip temperatures close to their safety thresholds to maximize energy savings within the liquid cooling system, without considering the potential negative impacts of elevated temperatures on the overall data center power consumption.

Ma et al. [19,20] investigated the use of artificial neural network models for control optimization of liquid cooling systems. They developed models to estimate coolant return temperatures [19] and the energy consumption of the liquid cooling system [20], both yielding satisfactory prediction accuracy. However, similar to previous studies, their modeling scope was limited to the liquid cooling system. Brewer et al. [21] proposed an open-source digital twin framework for liquid-cooled supercomputers, incorporating an analytical power and thermal model of the liquid cooling system. The model computes various outputs, including supply and return temperatures across different loops, as well as the power consumption of pumps and cooling tower fans. In addition to the liquid cooling system, the framework includes a power simulator that estimates IT power based on CPU and GPU utilization traces. However, it does not account for the temperature dependence of IT leakage power, which can significantly affect overall energy performance.

A notable attempt at more holistic modeling was presented by Stahlhut et al. [16], who developed a simulation model based on extensive experimental benchmark tests on ten liquid-cooled servers under varying coolant inlet temperatures and IT loads. Their model incorporated the liquid cooling system, heat transfer from IT equipment to server room air, and temperature-dependent IT leakage power. However, it did not consider the air cooling system responsible for removing heat from the server room environment. To the best knowledge of the authors, a holistic modeling approach capable of capturing these essential cross-system interactions remains a research gap. Consequently, the impact of supply water temperature on the total power consumption of data centers is not yet fully understood.

The main contributions of this study are as follows:

(1) A comprehensive analysis of 21 months of operational data from the direct liquid-cooled HAWK supercomputer at the High-Performance Computing Center Stuttgart (HLRS) [22] is conducted. To the authors' knowledge, this is the first study to systematically investigate the aforementioned cross-system interactions in a liquid-cooled data center based on long-term operational data.
(2) A data center–level power and thermal model is developed for the HAWK supercomputer. The model captures key dynamics, including the thermal behavior and energy performance of the liquid and air cooling systems, heat transfer from the IT equipment to the server room, and the temperature dependence of IT leakage power. The modeling approach presented in this study addresses the previously identified research gap in holistic modeling and is broadly applicable to other direct liquid-cooled data centers, providing a foundation for studies on data center–level performance analysis and operational optimization.
(3) An in-depth evaluation of the impact of supply water temperature on the energy performance of individual subsystems and the overall data center is conducted using the developed model.
(4) Insights that are broadly applicable to other liquid-cooled data centers are derived, providing guidance for power and thermal modeling, liquid cooling system control and operation, and the integration of heat reuse.

This paper is organized as follows. Section 2 provides an overview of the HAWK supercomputer and its cooling infrastructure. Section 3 presents an analysis of operational data from the HAWK supercomputer, highlighting the impact of supply water temperature on the performance of various systems. Section 4 describes the development of a data center-level power and thermal model. Based on this model, Section 5 presents simulation studies evaluating the impact of different supply water temperature setpoints on the energy performance of the liquid cooling system (Section 5.1), the air cooling system (Section 5.2), and the overall data center (Section 5.3). Section 6 discusses the generalizability and implications of the findings, and identifies limitations and directions for future work. Finally, Section 7 provides the conclusions.

2. HAWK supercomputer description

This section provides an overview of the HAWK supercomputer and its cooling infrastructure [22]. HAWK was operational from 2020 until its decommissioning in 2025. The system consists of 5632 nodes, each equipped with two 64-core AMD EPYC 7742 CPUs (thermal design power 225 W [23]) and 1.44 petabytes of primary memory. These nodes are arranged in 44 non-standard racks. The maximum power consumption for the entire system amounts to approximately 3.5 MW during normal operation.

HAWK employs a liquid cooling system (see Fig. 1) using direct liquid cooling with single-phase cold plates for CPU cooling. While most of the heat is removed by the liquid cooling system, a portion of the heat is passively transferred to the server room through convection and radiation. To maintain a specified temperature and humidity range in

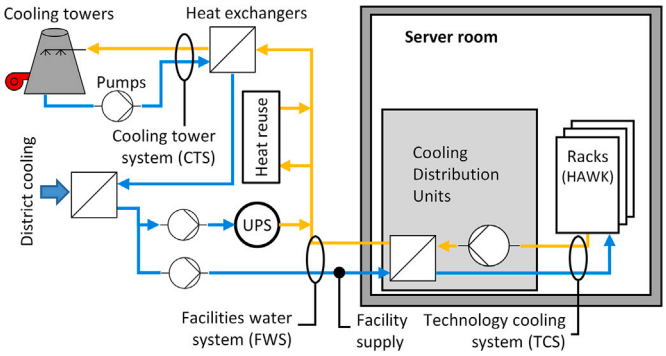


Fig. 1. Simplified schematic of the HAWK's liquid cooling system.

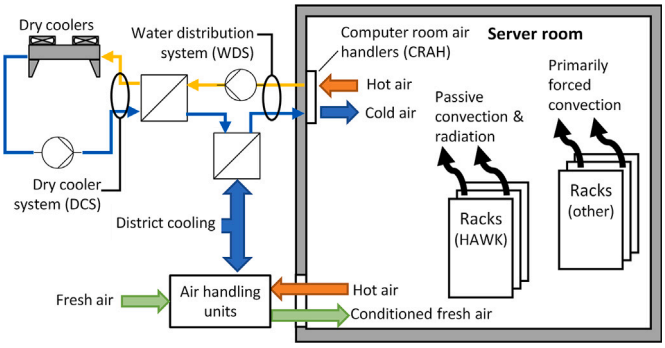


Fig. 2. Simplified schematic of the air cooling system at HLRS.

the server room, an air cooling system (see Fig. 2) is also employed. The system configuration of the HAWK supercomputer is shown in Table 1.

2.1. Liquid cooling system

As shown in Fig. 1, the facilities water system (FWS) transports water cooled by the cooling tower system (CTS), located in a separate building, to the technology cooling system (TCS), which directly cools the IT equipment. The cooled water is then directed to the heat exchangers within six cooling distribution units (CDUs) located adjacent to the server racks. These CDUs facilitate internal liquid circulation within the TCS, providing direct cooling for CPUs, dual inline memory modules (DIMMs), voltage regulators (VRs), chipsets, and internal network components. In addition to the CDUs, another heat source within the FWS is the flywheel uninterruptible power supply (UPS) units. These units counteract power frequency and voltage fluctuations, as well as brief interruptions in the power supply, but also generate considerable heat and thus require cooling.

Cooling for both the CDUs and the UPS units within the FWS primarily relies on four identical cooling towers in the CTS. These cooling towers are direct-contact cooling towers, where water is directly exposed to the cooling atmosphere, thereby transferring the source heat load directly to the air mainly through evaporation. The cooling tower fans are regulated via a PID controller to keep the facility supply temperature at its setpoint. Together, the cooling towers provide a nominal cooling capacity of 4.8 MW and have a total rated fan power of 134.4 kW.

During period of high outdoor temperatures or when the cooling towers are under maintenance, a district cooling network supplies 6 °C chilled water to the liquid cooling system. This chilled water is provided by vapor-compression chillers operated by the University of Stuttgart. The capacity of both the cooling towers and the district cooling is sufficient to independently manage the cooling load. Additionally, the FWS serves as a heat source for two heat pumps, which provide space heating to both the data center building and an adjacent office building. The cooling provided by these heat pumps is minimal, accounting for less than 1 % of the total heat produced annually [24].

Fig. 3 presents a more detailed overview of the connection between CDUs and the server racks. Each CDU is connected via piping to the racks that house the IT equipment. Within the TCS, coolant PG-25, a mixture of propylene glycol (25 %) and water, circulates to remove the heat from the IT equipment. The cooling of the IT equipment is achieved through the liquid cooling modules (LCM), with each LCM responsible for cooling two nodes. These LCMs use direct component liquid cooling and consist of cold plates, heat pipes, and cooling brackets.

2.2. Air cooling system

While most of the heat produced by the IT equipment is removed by the LCMs, a portion of the heat is passively dissipated into the server room via convection and radiation. In addition, the server room contains other server racks with a total cooling load of approximately 600 kW that rely exclusively on air cooling. To maintain a controlled environment, an air cooling system (see Fig. 2) is installed, which regulates temperature and humidity, and supplies fresh air.

Temperature control within the server room is managed by 20 computer room air handler (CRAH) units. The cold air supplied by these units absorbs the heat from the servers and returns to the CRAH units, where the transferred heat is rejected via an air–water heat exchanger to the water distribution system. When operating at full capacity, the CRAH units achieve an air exchange rate exceeding 140 per hour. The water distribution system supplies water to the CRAH units at a supply temperature of 14 °C.

During periods of low outdoor temperatures, heat removal from the water distribution system primarily relies on four dry coolers, each with a nominal cooling capacity of 1 MW and a total fan rated power of 30.24 kW. During periods of high outdoor temperature or when the dry coolers are under maintenance, cooling is supplemented by the district cooling.

In addition to temperature control, humidity regulation and fresh air supply are achieved via two parallel-connected air handling units (AHUs), providing 9000 m^3/h of conditioned outdoor air, equivalent to an air exchange rate of approximately three times per hour. To maintain humidity within specified limits, recirculated air can be mixed with outdoor air, and humidity is controlled through cooling coils (connected to the district cooling) for dehumidification, and air washers for humidification.

3. Measurement data analysis

The liquid cooling system of the HAWK supercomputer was operated under various facility supply temperature setpoints, providing a valuable dataset for analyzing the interactions among the key subsystems, including the liquid cooling system, the air cooling system, and the IT equipment. This measurement data analysis enables an in-depth examination of the cross-system interactions and establishes a robust foundation for developing a holistic modeling approach in the following section.

The building management system (BMS) at HLRS monitors and records the operational data of the liquid cooling system for the HAWK supercomputer at 5-min intervals. This includes temperature and flow rates across various system components, as well as critical electrical data such as the power inlet of the UPS units and the power supplied to the server racks from the UPS units.

Additionally, the power consumption of the facility room housing the liquid cooling system equipment is recorded. This includes not only the liquid cooling system elements, such as cooling tower fans, pumps, and heat pumps, but also auxiliary components including control valves, fan coil units for air conditioning, lighting, and power sockets within the facility room. However, the power consumption attributed to these auxiliary components is comparatively minimal.

The chillers in the district cooling network are located outside of HLRS and are therefore not monitored by the BMS. Instead, the power consumption of the district cooling system is estimated based on the measured cooling energy. According to data from the district cooling operator, a constant year-round energy efficiency ratio (EER) of 4 is assumed.

For this study, data spanning from December 31, 2021 to September 30, 2023 (184,008 data points recorded at 5-min intervals) were collected from the BMS. Throughout this period, the system was operated under various facility supply temperature settings. Initially, the facility supply temperature setpoint was fixed at 17 °C for approximately seven months (December 31, 2021 to July 19, 2022). It was then adjusted to a fixed setpoint of 20 °C, and finally transitioned to a dynamically controlled facility supply temperature setpoint (November 29, 2022 to September 30, 2023) ranging between 14 °C and 25 °C, set 6.5 K above the measured outdoor wet-bulb temperature. To evaluate the impact

Table 1
System configuration of the HAWK supercomputer.

Component	Parameter	Value
Server configurations		
IT equipment	Maximum power consumption (normal operation)	3.5 kW
	Maximum power consumption (LinPack operation)	4.1 kW
	Idle power consumption	1.6 kW
Liquid cooling system configurations		
Cooling tower	Quantity	4
	Nominal cooling capacity (per tower)	1.2 MW
	Number of fans (per tower)	2
	Fan rated power	16.8 kW each
Pump – CTS	Design flow rate	413.9 m^3/h
	Head	25.0 m
	Rated power	35.7 kW
Pump – FWS	Design flow rate	431.0 m^3/h
	Head	36.0 m
	Rated power	50.0 kW
Heat exchanger – CTS/FWS	Nominal overall heat transfer coefficient	2.4 MW/K
Cooling distribution unit	Quantity	6
	Design flow rate (per unit)	72.7 m^3/h
	Pump rated power (per unit)	12 kW
	Nominal overall heat transfer coefficient (per unit)	0.3 MW/K
Air cooling system configurations		
Server room	Total volume	3000 m^3
	Volume of space housing HAWK (estimated)	1000 m^3
CRAH	Total quantity	20
	Quantity serving HAWK (estimated)	6
	Server room temperature setpoint	24 °C
	Rated airflow (per unit)	22,000m^3/h
	Rated power (per unit)	4.8 kW
	Supply water temperature	14 °C
	Supply water flow rate (per unit)	13.0 m^3/h
	Nominal overall heat transfer coefficient (per unit)	12.3 kW/K
Air handling unit	Fresh air supply (total)	9000m^3/h
	Fresh air supply (for HAWK)	3000m^3/h
	Server room relative humidity range	20–50 %
	Rated fan power	1.35 kW
Pump – WDS	Design flow rate	259.0 m^3/h
	Rated power	29.5 kW
Pump – DCS	Design flow rate	215.0 m^3/h
	Rated power	22.0 kW
Dry cooler	Quantity	4
	Nominal cooling capacity (per tower)	250.0 kW
	Number of fans (per tower)	12
	Fan rated power	630 W each
Heat exchanger – DCS/WDS	Nominal overall heat transfer coefficient	625.2 kW/K
District cooling configurations		
Supply temperature		6 °C
Energy efficiency ratio (estimated)		4

of facility supply temperature, the period with a fixed facility supply temperature of 17 °C and the period under dynamic temperature control were selected for comparative analysis. To account for system inertia and hydraulic response time, the data was resampled into one-hour interval by taking mean values.

Fig. 4 compares the system performance under these two different facility supply temperature settings. As shown in the left plot, at lower outdoor wet-bulb temperatures (below 5 °C), both temperature settings result in an inverse energy efficiency ratio (IEER) of approximately 0.05 for the liquid cooling system. The IEER, defined as the ratio of power consumption to the cooling load, indicates that the liquid cooling system requires power equivalent to about 5 % of the cooling load to dissipate it into the outdoor environment. As the wet-bulb temperature increases, the IEER increases in both settings, although at different rates. Initially, both settings demonstrate similar performance, reaching an IEER of approximately 0.08 at a wet-bulb temperature of around 10 °C, where the facility supply temperatures are comparable (17 °C for the fixed setting and 16.5 °C for the dynamic setting). This increase in the ratio is mainly due to the increased fan speeds in the cooling towers, which approach their maximum cooling capacity at the given setpoint. Beyond this wet-bulb temperature, the performance of the two settings begins to diverge significantly. Under the fixed 17 °C setpoint, the cooling towers reach their operational limits, requiring the activation of the district cooling system. Since the district cooling system is less efficient compared to the cooling tower system, the IEER increases sharply. For instance, at a wet-bulb temperature of 19 °C, district cooling covers 50.3 % of the cooling load, resulting in an IEER of 0.21. In contrast, under the same outdoor conditions, the dynamic setting allows the facility supply temperature to reach the upper limit of 25 °C, enabling the cooling towers to handle 98.2 % of the cooling load. As a result, the

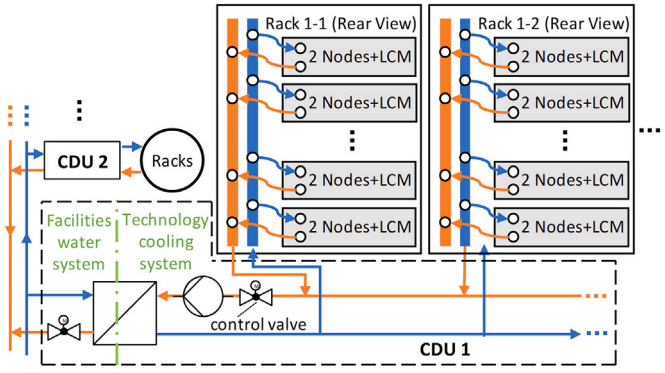


Fig. 3. Simplified schematic of the the connection between CDUs and the server racks in the HAWK supercomputer.

IEER remains significantly lower at 0.077, which is 63.3 % less than that of the fixed setting.

The rise in temperature in the liquid cooling system significantly enhances its energy efficiency and reduces the demand for district cooling. However, it also increases the temperature of IT equipment within the server, which presents certain drawbacks. Firstly, a higher IT equipment temperature results in a greater temperature difference between the IT equipment and the surrounding server room air. This greater temperature difference leads to an increase in heat transfer from IT equipment to server room air, as shown in the middle plot. Such increased heat dissipation to the server room air is unfavorable, as air cooling systems are generally less efficient than liquid cooling systems. The setting with a fixed supply temperature of 17 °C shows no clear correlation between the heat dissipated to the air (expressed as a percentage of the total heat generated by the IT equipment) and the outdoor wet-bulb temperature. This ratio consistently remains between 2.1–2.7 %. In contrast, under the dynamic setting, this ratio increases approximately linearly with rising wet-bulb temperature. Specifically, the proportion of heat dissipated to the server room air increases from approximately 2.6 % at a wet-bulb temperature of 8.5 °C to 5.0 % at 18.5 °C, nearly doubling.

Another drawback of increasing the IT equipment temperature is the rise in leakage power in semiconductors, which leads to an increase in IT power consumption. Measurement results (right plot) indicate that this increase can be approximated by a linear function, although the data exhibit considerable uncertainty. According to feedback from HLRS, the HAWK supercomputer generally operates at a consistent utilization rate, indicating that the observed uncertainties are primarily due to variations in the software and workload running on the servers. Specifically, increasing the facility supply temperature from 17 °C to 25 °C results in an increase in IT power consumption of approximately 87.6 kW (3.15 %).

4. Power and thermal model for HAWK supercomputer

Based on the results obtained from the measurement data analysis presented in Section 3, it has become evident that modeling only the liquid cooling system is insufficient for developing effective control strategies. The liquid cooling system's operation considerably impacts both the IT equipment's power consumption and heat transfer to the server room air, and consequently, the power consumption of the air cooling system. In order to capture all essential dynamics associated with the liquid cooling system's operation, a holistic power and thermal model for the HAWK supercomputer is developed in this section.

Fig. 5 provides a simplified depiction of the holistic model, comprising three interconnected submodels: the liquid cooling system model, the air cooling system model, and the server and UPS system model.

The liquid cooling system model calculates various outputs, including inlet and outlet temperatures at each component, thermal performance and power consumption of essential components, as well as fresh water usage from cooling towers. The calculations are based on inputs including weather data, heat reuse rate, the number of active cooling towers, heat flows from IT equipment and UPS units, the facility supply temperature setpoint, and setpoints for flow rates in different loops.

The air cooling system model calculates the power consumption required to regulate the server room environment, based on inputs such as heat flow from IT equipment to server room air, weather data, and the desired server room temperature setpoint.

The liquid cooling module model predicts the heat dissipation from IT equipment to both the liquid cooling system and the server room air. It also calculates the average and maximum CPU case temperatures. The IT equipment model and UPS model predict the power consumption of the IT equipment and UPS units. Inputs to these models include server utilization rates, supply temperature and mass flow rate within the TCS ($T_{TCS,supply}$ and $\dot{m}_{TCS}$), and server room temperature (T_{room}).

Due to the extensive measurement data available, the essential dynamics of most system components could be accurately identified

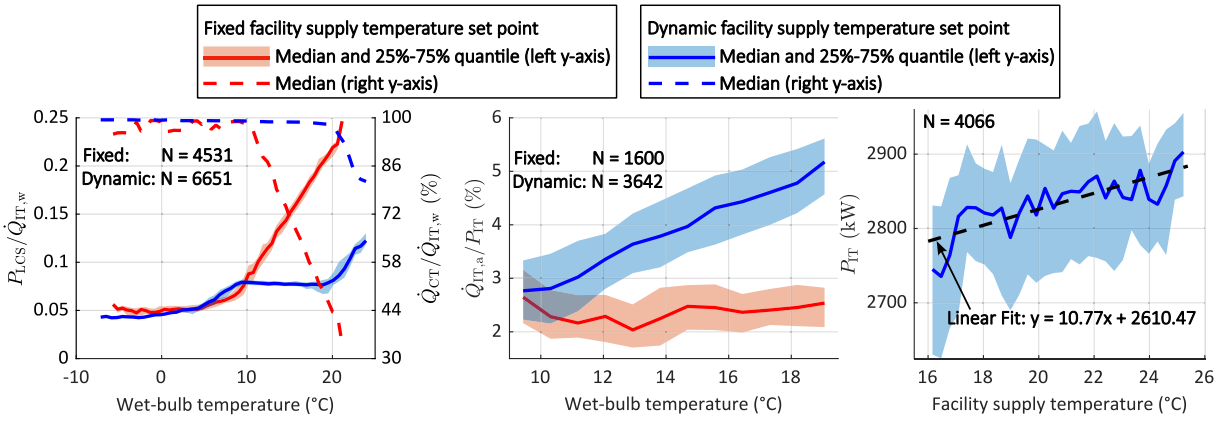


Fig. 4. Comparison of measurement data from periods with fixed and dynamic facility supply temperature settings. P_{LCS} denotes the power consumption of the liquid cooling system, and P_{IT} represents the total power consumption of the IT equipment. $\dot{Q}_{IT,w}$ and $\dot{Q}_{IT,a}$ refer to the heat transferred from the IT equipment to the liquid and air cooling systems, respectively. $\dot{Q}_{CT}$ indicates the cooling capacity of the cooling towers. Data points with measurement anomalies were excluded. To isolate the effect of the facility supply temperature, only data points with FWS water mass flow rates between 87.5–92.5 kg/s and periods when servers were not idle were selected. For the left plot, data points from periods when cooling towers were deactivated were excluded. For the middle and right plots, periods when the UPS units were deactivated were excluded, as P_{IT} corresponds to the power measured at the UPS output.

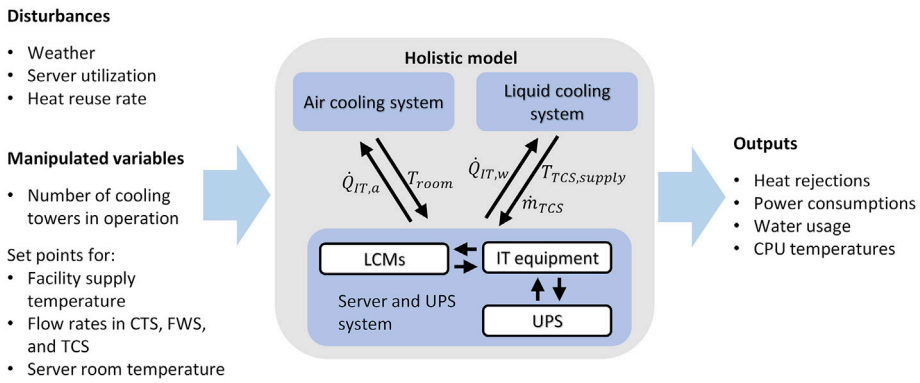


Fig. 5. Overview of the holistic power and thermal model for HAWK supercomputer.

and modeled. For components without measurement data, the models were developed using detailed information from product catalogs, planning and commissioning documentation. The following subsections describe the modeling approaches and methodologies employed for each essential component of the submodels.

4.1. Liquid cooling module model

The liquid cooling module model consists of two parts. The first part estimates the heat transferred from the IT equipment to the server room, while the second estimates CPU case temperatures. Both parts are described in the following subsections.

4.1.1. Heat transfer from IT equipment to the server room

Fig. 6 illustrates the heat transfer process from IT equipment to the server room environment. To simplify the modeling of LCMs, individual IT components are not viewed as separate units. Instead, they are collectively represented as a unified heat source. Heat transfer from IT equipment to the server room includes the following processes: first, the IT equipment generates heat, which is conducted to the directly attached elements, such as the substrate and the LCMs. Heat is then transferred from the IT equipment and these heated elements to the air inside the server rack via convection, and to the rack envelope via radiation. Finally, the server rack transfers the heat to the server room through both convection and radiation.

Based on this understanding, a thermal resistance network is constructed, as depicted on the right side of Fig. 6. This model calculates the heat transfer from IT equipment to server room ($\dot{Q}_{IT,a}$) and to the coolant in the LCMs ($\dot{Q}_{IT,w}$) using the following equations:

$$\dot{Q}_{IT,a} = \frac{T_{LCM} - T_{rack}}{R_{LCM,rack}} + \frac{T_{IT} - T_{rack}}{R_{IT,rack}} = \frac{T_{rack} - T_{room}}{R_{rack,room}} \tag{1}$$

$$\dot{Q}_{IT,w} = P_{IT} - \dot{Q}_{IT,a} = \frac{T_{IT} - T_{LCM}}{R_{IT,LCM}} - \frac{T_{LCM} - T_{rack}}{R_{LCM,rack}} \tag{2}$$

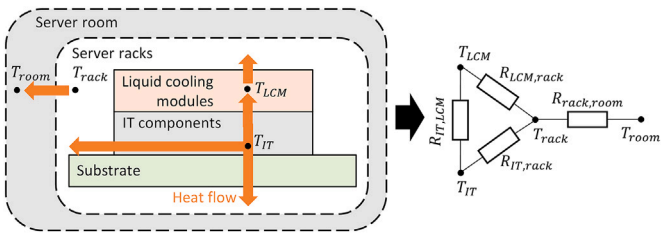


Fig. 6. Schematic representation of the heat transfer process from IT equipment to the server room environment (left) and the corresponding thermal resistance network model (right).

Here, T_{IT} is the average temperature of the IT equipment. The mean coolant temperature in the LCMs, T_{LCM}, is approximated as the average of the TCS supply and return temperatures. T_{rack} represents the combined air and surface temperature of the server rack, and T_{room} is the server room temperature. The symbol R denotes the total thermal resistance between the respective elements.

By rearranging Eqs. (1) and (2), the expression for $\dot{Q}_{IT,a}$ can be formulated as a function of P_{IT}, T_{LCM}, T_{room}, and the relevant thermal resistances:

$$\begin{aligned}\dot{Q}_{IT,a} = &\Big(R_{IT,LCM}R_{LCM,rack}P_{IT} + (R_{IT,LCM} + R_{LCM,rack} \\ &+R_{IT,rack})T_{LCM} - (R_{IT,LCM} + R_{LCM,rack} + R_{IT,rack})T_{room}\Big) \\ &\Big/\Big(R_{IT,LCM}R_{LCM,rack} + R_{IT,LCM}R_{rack,room} + R_{IT,rack}R_{LCM,rack} \\ &+R_{IT,rack}R_{rack,room} + R_{LCM,rack}R_{rack,room}\Big)\end{aligned} \tag{3}$$

Several variables in Eq. (3) can reasonably assumed to be constant. The thermal resistance $R_{IT,LCM}$ depends on the coolant flow rate in the LCMs, which operates at a fixed setpoint during HAWK's operation. Therefore, $R_{IT,LCM}$ is treated as constant. The convective and radiative heat transfer coefficients between different elements, as well as the effective surface areas of those elements for heat transfer, are assumed to be constant. Therefore, $R_{IT,rack}$, $R_{LCM,rack}$, and $R_{rack,room}$ are also considered constant. With these assumptions, Eq. (3) simplifies to the following form:

$$\dot{Q}_{IT,a} = \beta_{lcm,0}P_{IT} + \beta_{lcm,1}(T_{LCM} - T_{room}) \tag{4}$$

where, $\beta_{lcm,0}$ and $\beta_{lcm,1}$ are model parameters.

Fig. 7 shows the fitted linear regression model for estimating $\dot{Q}_{IT,a}$, as formulated in Eq. (4). Since the server room is maintained at a nearly constant air temperature, T_{room} is treated as constant during model fitting. The model demonstrates good agreement with the reference data points, achieving an R^2 value of 0.69, indicating that a majority portion of the variance is explained by the fitted model. The CVRMSE value is 23.54 %, which is relatively high compared to the near-zero NMBE value. This can be attributed to measurement uncertainties: $\dot{Q}_{IT,a}$ is not directly measured but calculated as the difference between measured P_{IT} and $\dot{Q}_{IT,w}$. Given that $\dot{Q}_{IT,a}$ makes up only a small fraction of P_{IT}, even minor measurement errors can lead to relatively large deviations in the calculated values.

4.1.2. CPU case temperature estimation

To determine the maximum and average CPU case temperatures, a thermal model representing the cooling circuit within the LCM is developed (see Fig. 8). Within each LCM, the coolant first passes through DIMMs, VRs, and internal network components before reaching the

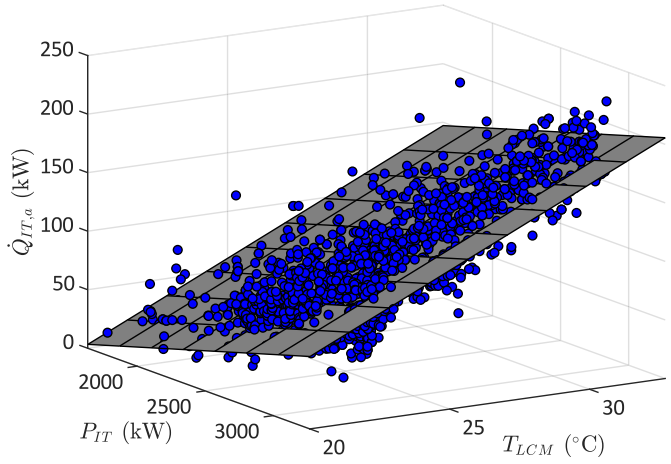


Fig. 7. Fitted linear regression model (the plane) with reference data points. A subset of 2000 data points is randomly selected from the full dataset for visualization. Measurement data covers the period from April 9, 2022 to October 9, 2023. Total number of data points: N = 10,092, sampling interval: 1 h. Model performance metrics: R-Squared (R^2) = 0.69, root mean squared error (RMSE) = 23.76 kW, coefficient of variation of the RMSE (CVRMSE) = 23.54 %, mean bias error (MBE) = 3.79×10^{-13} kW, normalized MBE (NMBE) = 3.75×10^{-13} %.

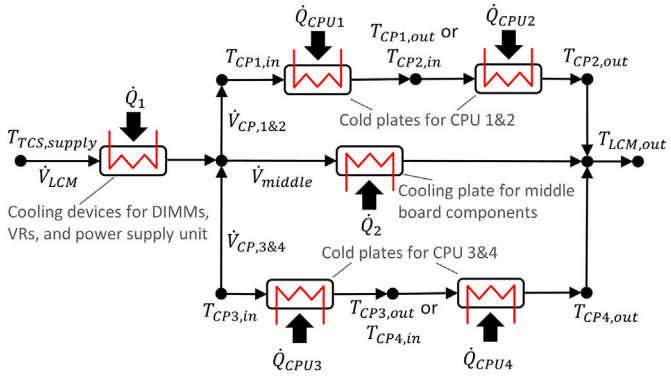


Fig. 8. Schematic diagram of the thermal model for the cooling circuit within a liquid cooling module.

CPUs. As a result, the coolant inlet temperatures to the upstream CPU cold plates ($T_{CP1,in}$ and $T_{CP3,in}$) are higher than the initial coolant inlet temperature to the LCM ($T_{TCS,supply}$). Based on the temperature distribution calculated throughout this cooling circuit, the CPU case temperature (T_{case}) is estimated using the following relationship:

$$T_{case} = R_{CP}\dot{Q}_{CPU} + T_{CP} \tag{5}$$

Here, $\dot{Q}_{CPU}$ denotes the heat transfer rate from the CPU to the cold plate, and T_{CP} represents the mean coolant temperature within the cold plate. The thermal resistance of the CPU cold plate, R_{CP}, depends on the coolant volumetric flow rate through the cold plate ($\dot{V}_{CP}$). The relationship between thermal resistance and coolant flow rate for both upstream and downstream cold plates is derived from data provided by the manufacturer Hewlett Packard Enterprise [25]. Furthermore, the manufacturer specifies the distribution of coolant flow rates among the three parallel streams within the LCM.

For estimating the maximum and average CPU case temperatures, certain assumptions and simplifications are required. First, it is assumed that 100 % of the IT power is transferred to the coolant. To calculate the maximum CPU case temperature, it is assumed that whenever the server is active, all IT components within at least one LCM operate at their maximum power consumption, regardless of the overall IT power consumption. For the average CPU case temperature estimation, an even distribution of IT power consumption across all server nodes is assumed.

4.2. IT equipment and UPS model

The power consumption of the CPU (P_{CPU}) can be expressed as the sum of idle ($P_{CPU,idle}$), dynamic ($P_{CPU,dynamic}$), and static ($P_{CPU,static}$) power components [26]:

$$P_{CPU} = P_{CPU,idle} + P_{CPU,dynamic} + P_{CPU,static} \tag{6}$$

The idle power is considered constant:

$$P_{CPU,idle} = \alpha_0 \tag{7}$$

The dynamic power consumption of a CPU is determined by the effective switching capacitance (c_e), the supply voltage (E_{dd}), and the clock frequency (f). Since the clock frequency is proportional to the CPU utilization (u_{CPU}), the dynamic power can be expressed as follows, if the c_e and E_{dd} are constant [27]:

$$P_{CPU,dynamic} = \frac{1}{2}c_e E_{dd}^2 \, f = \alpha_1 u_{CPU} \tag{8}$$

Static power arises from leakage currents and depends on the CPU die temperature. In this study, it is approximated as a linear function of the CPU case temperature (T_{case}):

$$P_{CPU,static} = \alpha_2 T_{case} \tag{9}$$

As the CPUs account for the majority of the IT equipment's total power consumption, the overall IT equipment power (P_{IT}) is approximated as proportional to the CPU power consumption:

$$P_{IT} = \beta_{it,0} + \beta_{it,1}\bar{u}_{CPU} + \beta_{it,2}\bar{T}_{case} \tag{10}$$

where $\bar{u}_{CPU}$ and $\bar{T}_{case}$ are the average CPU utilization and average case temperature, respectively. $\beta_{it,0}$, $\beta_{it,1}$, and $\beta_{it,2}$ are model parameters.

Since direct measurements of the CPU case temperature were not available for this study, it was calculated using the measured facility supply temperature and the models described in Sections 4.1.2 and 4.3. Additionally, although direct measurements of CPU utilization were not accessible, feedback from HLRS indicates that the HAWK supercomputer generally operates at a consistent utilization rate. Consequently, the parameter $\beta_{it,2}$ in the IT power model, which characterizes the dependence of IT power on CPU case temperature, was identified using the measured overall IT equipment power, the calculated average CPU case temperature, and an assumed constant CPU utilization rate (see Fig. 9).

To enable future analyses involving varying utilization profiles, the idle power component ($\beta_{it,0}$) and the utilization-dependent coefficient ($\beta_{it,1}$) in Eq. (10) were further estimated based on known reference points: at a facility supply temperature of 25 °C, the total IT power is approximately 3500 kW at 100 % utilization and 1600 kW at idle (see Table 1).

For the UPS units, the power loss is modeled as a third-order polynomial function of P_{IT}:

$$P_{UPS,loss} = \beta_{ups,0} + \beta_{ups,1}P_{IT} + \beta_{ups,2}P_{IT}^2 + \beta_{ups,3}P_{IT}^3 \tag{11}$$

where, $\beta_{ups,0}$, $\beta_{ups,1}$, $\beta_{ups,2}$, and $\beta_{ups,3}$ are model parameters. In this study, these model parameters are identified using measurement data.

4.3. Liquid cooling system model

The liquid cooling system model provides a detailed representation of all critical components and their network within the system. It specifically includes elements such as cooling towers and their fans, heat exchangers, and pumps.

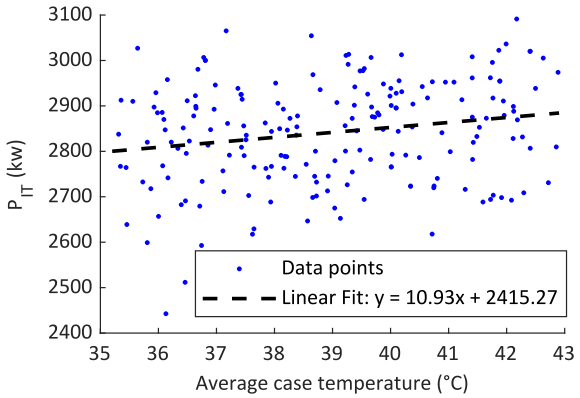


Fig. 9. Parameter identification of $\beta_{it,2}$ in the IT power model ($\beta_{it,2} = 10.93$). A subset of 200 data points is randomly selected from the full dataset for visualization. Sampling interval: 1 h, total number of data points: $N = 3{,}228$. Model performance metrics: RMSE = 130.01 kW, CVRMSE = 4.61 %, MBE = 18.38 kW, NMBE = 0.65 %.

The heat dissipation rate of the cooling towers ($\dot{Q}_{CT}$), is calculated using the explicit model [28] outlined in Eq. (12). The model was fitted using measurement data collected over a 21-month period. As shown in Fig. 10, the fitted model demonstrates good agreement with the data, achieving a CVRMSE value of 3.68 % and an NMBE value of 0.05 %. Based on the derived model, the calculated heat dissipation rate of the cooling towers at nominal operational conditions and a wet-bulb temperature of 5 °C is calculated to be 4104.12 kW, 14.5 % lower than the value listed in the product catalog (4800 kW). This discrepancy highlights the importance of incorporating measurement data into the modeling process rather than relying exclusively on product catalog specifications.

The water loss due to evaporation in the cooling towers is estimated under the assumption that the outlet air is fully saturated. If the enthalpy change of the water is insufficient to fully saturate air at its inlet dry-bulb temperature, the air dry-bulb temperature is assumed to remain constant at its inlet value, and the humidity ratio is increased until the enthalpy changes of air and water match.

$$\dot{Q}_{CT} = \frac{\dot{m}_a}{1+\beta_{ct,0}\phi_{aw}}[\beta_{ct,1}\left(T_{CT,i}+WBT\right)\left(T_{CT,i}-WBT\right) + \beta_{ct,2}\left(T_{CT,i}-WBT\right)+\beta_{ct,3}] \tag{12}$$

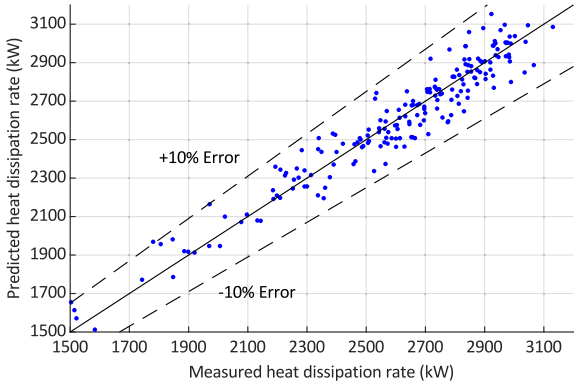


Fig. 10. Comparison between predicted and measured heat dissipation rate of the cooling towers ($\dot{Q}_{CT}$). A subset of 200 data points is randomly selected from the full dataset for visualization. Measurement data covers the period from December 31, 2021 to September 30, 2023. Sampling interval: 1 h, total number of data points: $N = 14{,}405$. Model performance metrics: RMSE = 95.52 kW, CVRMSE = 3.68 %, MBE = 1.37 kW, NMBE = 0.05 %.

Here, $\dot{m}_a$ is the air mass flow rate entering the cooling tower, ϕ_{aw} denotes the air-to-water mass flow rate ratio, and WBT represents the outdoor wet-bulb temperature. $\beta_{ct,0}$, $\beta_{ct,1}$, $\beta_{ct,2}$, and $\beta_{ct,3}$ are model parameters.

All heat exchangers within the liquid cooling system are insulated counter-flow plate heat exchangers. For simplicity, the thermal capacitance of the heat exchangers is neglected in the model. Heat transfer in the exchangers is modeled using the effectiveness-NTU method [29], with the overall heat transfer coefficient (UA) modeled as dependent on flow rates rather than constant.

As shown in Eq. (13), the total thermal resistance (R_{tot}) between hot and cold fluids, defined as the inverse of the overall heat transfer coefficient of the heat exchanger (UA), comprises the conductive thermal resistance of the heat exchanger material (R_{cond}) and the convective thermal resistances on the hot and cold fluid sides ($1/h_hA_h$ and $1/h_cA_c$). The convective heat transfer coefficients (h) are determined from the Nusselt number ($Nu = hL/k$, where L is the characteristic length, and k is the thermal conductivity of the fluid), which, in this model, is calculated using the Dittus-Boelter correlation [30] given in Eq. (14). The Reynolds number ($Re = uL/v$) links the fluid velocity (u) to the Nusselt number. Since fluid velocity is directly proportional to volumetric flow rate ($\dot{V}$), changes in flow rates directly influence the convective heat transfer.

The R_{tot}, the effective convective heat exchange areas on both sides (A_h and A_c), and the characteristic length (L) are constant and defined by the heat exchanger geometry. Fluid properties such as the Prandtl number (Pr) and kinematic viscosity (v), although temperature- and pressure-dependent, are simplified here as constants. Consequently, fluid volumetric flow rates remain the single variable influencing R_{tot}. Integrating these assumptions and relationships, R_{tot} is reformulated as shown in Eq. (15), with $\beta_{hx,0}$, $\beta_{hx,1}$, and $\beta_{hx,2}$ as model parameters.

$$R_{tot} = \frac{1}{UA} = \frac{1}{h_hA_h} + R_{cond} + \frac{1}{h_cA_c} \tag{13}$$

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{b}, \quad b = \begin{cases} 0.4 & \text{(cold fluid)} \\ 0.3 & \text{(hot fluid)} \end{cases} \tag{14}$$

$$R_{tot} = \frac{1}{UA} = \frac{1}{\beta_{hx,0}\dot{V}_h^{0.8}} + \beta_{hx,1} + \frac{1}{\beta_{hx,2}\dot{V}_c^{0.8}} \tag{15}$$

Fig. 11 illustrates an example of model fitting for the heat exchanger between the CTS and FWS. The results show a good agreement between the model and the measurements, with an RMSE value of 46.44 kW and a CVRMSE value of 4.41 %. Additionally, the measurement data show that both the CTS and FWS operate at flow rates below their nominal values. As a result, 90 % of the observed UA values fall within the range of 939–1166 kW/K. Based on the derived model, the calculated nominal UA at nominal flow rates is approximately 1474 kW/K, significantly lower than the 2400 kW/K value listed in the product catalog. This effect can primarily be attributed to fouling, which degrades the heat transfer performance of heat exchangers over time [31]. This discrepancy again underscores the necessity of incorporating measurement data into the modeling process. Furthermore, ignoring the flow-dependent variability of UA could lead to an overestimation of cooling capacity, which could increase the risk of overheating or operational failures in model-based control scenarios.

All essential piping within the cooling system is included in the model, but is implemented as adiabatic since the pipes are insulated. Measurement data support the assumption that heat exchange through insulated piping has a negligible impact on the overall energy balance of this liquid cooling system [24].

The total power consumption of the liquid cooling system is modeled as follows:

$$P_{LCS} = P_{CTS,fans} + P_{CTS,pumps} + P_{FWS,pumps} + P_{TCS,pumps} + P_{DC} \tag{16}$$

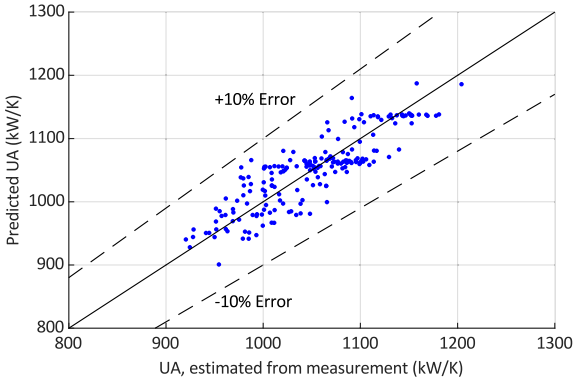


Fig. 11. Comparison between predicted UA values and UA values estimated from measurements. A subset of 200 data points is randomly selected from the full dataset for visualization. Measurement data covers the period from January 1 to October 7, 2023. Sampling interval: 1 h, total number of data points: N = 5,419. Model performance metrics: RMSE = 32.25 kW, CVRMSE = 3.06 %, MBE = −0.72 kW, NMBE = −0.07 %.

where $P_{CTS,fan}$ is the power consumption of the cooling tower fans, and $P_{CTS,pump}$, $P_{FWS,pump}$, and $P_{TCS,pump}$ represent the power consumption of pumps in the CTS, FWS, and TCS loops, respectively. P_{DC} denotes the power consumption of the district cooling system, which is estimated based on its cooling capacity using a constant EER of 4.

The power consumption of each cooling tower fan (P_{fan}) is modeled as a power function of the fan speed (n_{fan}). The model parameters, $\beta_{fan,0}$ and $\beta_{fan,1}$, were identified using measurement data:

$$P_{fan} = \beta_{fan,0} \cdot n_{fan}^{\beta_{fan,1}} \tag{17}$$

The power consumption of all pumps is explicitly represented in the model. For simplicity, pumps configured in parallel are combined into a single equivalent component. All pumps are speed-controlled, glanded pumps with negligible heat transfer to the liquid. Due to the absence of high-resolution measurement data for the power consumption of individual pump, data-based modeling approach was not feasible. Instead, pump power is calculated using a physics-based formulation [32]:

$$P_{pump} = \dot{V} \cdot \Delta p / \eta \tag{18}$$

where Δp is the pressure increase across the pump, and η is the pump efficiency. Δp is determined from the system curve, expressed as [32]:

$$\Delta p = \beta_{sys,0} + \beta_{sys,1} \cdot \dot{V}^2 \tag{19}$$

For closed-loop systems, the static head component $\beta_{sys,0}$ is zero. For open-loop systems such as the CTS, $\beta_{sys,0}$ is estimated based on the geometrical configuration of the cooling towers. $\beta_{sys,1}$ is derived from design flow rate and head information (see Table 1). η is represented as a third-order polynomial function of flow rate and pressure increase:

$$\eta = f(\dot{V}, \Delta p) \tag{20}$$

The coefficients of this polynomial were derived from pump product catalog data.

Since the liquid cooling system accounts for more than 95 % of the total cooling load, validation of the liquid cooling system model is essential to ensure accurate predictions of the cooling towers' cooling capacity and overall system power consumption. For model validation, measurement data, including cooling loads, outdoor conditions, mass flow rates in the different system loops, and facility supply temperature setpoint, were provided as model inputs.

Fig. 12 compares the measured and simulated monthly electric energy consumption of the liquid cooling system. During the winter months, the cooling towers and pumps were the primary power consumers. From May to August, power consumption increased significantly for both the cooling tower system and the district cooling system due to higher outdoor temperatures. In June, district cooling becomes the major power consumer, accounting for 162.24 MWh, or 55.10 % of the total electric energy consumption. Despite relatively cool weather in November, district cooling usage remains high. This is because the cooling tower system was deactivated for three days during that month. Overall, the simulation and measurement data exhibit a high level of agreement. Evaluation of hourly data shows a CVRMSE value of 11.52 % and a NMBE value of −1.09 %.

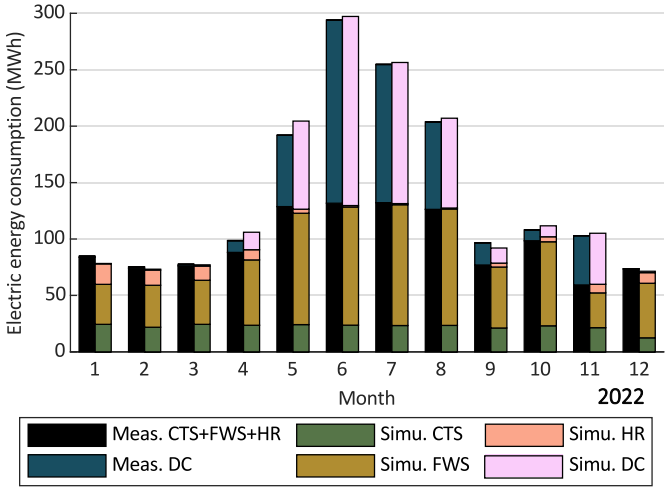


Fig. 12. Measured and simulated monthly electric energy consumption in liquid cooling system for year 2022 (for hourly data, RMSE: 21.86 kW, CVRMSE: 11.52 %, MBE: −2.08 kW, NMBE: −1.09 %). The electric energy consumption of the facility room, which houses the cooling towers, pumps in CTS and FWS, and heat pumps for heat reuse, is measured collectively. The electric energy consumption of the district cooling system is estimated using an EER of 4. HR: heat reuse, DC: district cooling, Meas.: measured, Simu.: simulated.

Fig. 13 shows the validation of the cooling towers' heat dissipation rate and the corresponding power consumption of the facility room, which houses the liquid cooling system equipment (including components in the CTS, FWS, and heat reuse system), over a five-day period. The left plot clearly demonstrates a reduction in cooling towers' cooling capacity during daytime hours due to higher outdoor temperatures, which necessitates the use of district cooling. Variations in the facility room's power consumption (as shown in the right plot) are primarily

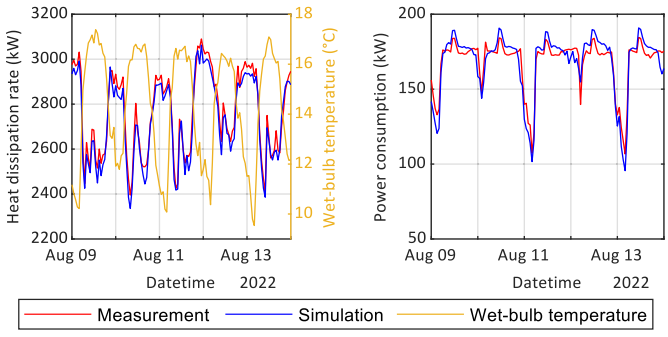


Fig. 13. Measured and simulated heat dissipation rate of the cooling towers (left plot) and power consumption of the liquid cooling system' facility room, including cooling towers, pumps in cooling tower system and facilities water system, and heat pumps for heat reuse (right plot), for the period August 9th to 14th, 2022. During this period, approximately 90 % of the cooling load remained within the range of 2855–3050 kW, with no correlation to the time of day.

attributed to changes in the cooling towers' fan speed. The simulation results closely align with the measured data, further confirming the accuracy of the developed liquid cooling system model.

4.4. Air cooling system model

The air cooling system model includes CRAH units, dry coolers, AHUs, pumps, and heat exchangers. The dry cooler system and the water distribution system are modeled according to their original system configuration. However, given that the HAWK supercomputer accounts for only a portion of the total air cooling load in the server room, only the server room space housing the HAWK servers, and the corresponding CRAH units and AHUs serving this space, are explicitly modeled. The cooling load from other servers and the heat transfer from outside through the building envelope are modeled as a linear function of the outdoor dry-bulb temperature, as suggested by measurement data.

The server room is modeled using energy and mass balances. Within the energy balance, the change in server room air temperature (dT_{room}) is determined by the heat dissipation from HAWK to the room air ($\dot{Q}_{IT,a}$), the cooling capacities provided by CRAH units ($\dot{Q}_{CRAH}$), and AHUs ($\dot{Q}_{AHU}$), as expressed by Eq. (21). In the mass balance, changes in the absolute humidity ratio within the server room are managed solely by the AHU units.

$$dT_{room} = \frac{t}{c_{p,a} \cdot V_{room} \cdot \rho_a} \left(\dot{Q}_{IT,a} - \dot{Q}_{CRAH} - \dot{Q}_{AHU} \right) \tag{21}$$

Here, t is time, $c_{p,a}$ represents the air specific heat capacity, ρ_a the air density, and V_{room} the volume of the server room housing the HAWK servers.

CRAH units are modeled as cross-flow heat exchangers with a mixed air stream on the hot side, using the effectiveness–NTU method. Their UA is simplified as constant and derived from the product catalog. Airflow through the CRAH units is controlled by a PID controller to maintain the server room temperature below the specified setpoint. AHUs are primarily responsible for humidity control, ensuring that server room humidity and dew point remain within the desired range. The AHU model includes fans, an air-to-water heat exchanger supplied by chilled water from the district cooling system, and an air washer fed by fresh water for humidification. A PID controller regulates the chilled water flow rate for dehumidification. Dry coolers are modeled as cross-flow heat exchangers with a mixed air stream on the cold side, using a constant UA value derived from the product catalog.

The total power consumption of the air cooling system is calculated as follows:

$$P_{ACS} = P_{dry,fans} + P_{dry,pumps} + P_{dis,pumps} + P_{CRAH,fans} + P_{AHU,fans} + P_{CRAH,DC} + P_{AHU,DC} \tag{22}$$

where $P_{dry,fans}$, $P_{CRAH,fans}$, and $P_{AHU,fans}$ denote the fan power consumption of the dry coolers, CRAH units, and AHUs, respectively. $P_{dry,pumps}$ represents pump power consumption within the dry cooler circuit, and $P_{dis,pumps}$ denotes the power consumption of circulation pumps distributing chilled water to the CRAH units. Additionally, $P_{CRAH,DC}$ and $P_{AHU,DC}$ represent the power consumption attributed to district cooling for CRAH units and AHUs, respectively.

The fan power consumption of the CRAH units and dry coolers is modeled using the same approach as for the cooling tower fans (see Eq. (13)), with model parameters fitted to data from the design documentation. The AHU fans, as well as the pumps in the dry cooler system and the water distribution system, operate at fixed speeds. Therefore, their power consumption is reasonably estimated as constant, based on values provided in the design documentation.

As the dry cooler system and water distribution system also serve cooling loads from other servers, power consumption components such as $P_{dry,fans}$, $P_{dry,pumps}$, $P_{dis,pumps}$, and $P_{CRAH,DC}$ must be proportionally allocated. In this model, their allocation to other servers is determined based on the ratio of their cooling load relative to the total, calculated under the condition that HAWK operates with a fixed facility supply temperature of 17 °C in the liquid cooling system.

5. Simulation analysis

Based on the holistic model developed in Section 4, simulations were conducted to analyze the impact of facility supply temperature on the power consumption and operational performance of the individual systems and components within the HAWK supercomputer. The simulations were conducted under varying outdoor conditions, using three facility supply temperature setpoints: 17 °C, 22 °C, and 27 °C. To isolate the effect of facility supply temperature, all other model inputs were fixed at their typical setpoints observed during real-world operation, and heat reuse was set to zero.

Table 2 summarizes the simulation results for the cooling loads from IT equipment and UPS units, along with their distribution between the liquid and air cooling systems under the three different temperature setpoints. As shown, the total cooling load increases by approximately 120.3 kW when the facility supply temperature rises from 17 °C to 27 °C. However, the cooling load managed by the liquid cooling system remains almost constant, increasing by only about 2.8 kW. Consequently, nearly the entire additional cooling load resulting from the higher facility supply temperature is dissipated via the air cooling system, leading to an increase in the air cooling share from 2.8 % to 6.8 %.

To avoid adverse impacts on computational performance, in our case, CPU case temperatures should remain below approximately 70 °C, as higher temperatures could trigger automatic reductions in CPU clock speed. The simulation results indicate that increasing the facility supply temperature from 17 °C to 27 °C leads to an increase in CPU case temperatures of approximately 10.8 K. This increase is slightly greater than the increase in the facility supply temperature due to the simultaneous increase in the power consumption of IT equipment. Nevertheless, even at a facility supply temperature of 27 °C, the CPU case temperatures remain well within acceptable limits. Specifically, the maximum CPU case temperature (for a node operating at full capacity) is approximately 50.4 °C, and the calculated average CPU case temperature used for IT power estimation is around 45.2 °C.

Further analysis of system performance under varying outdoor conditions is presented in the following subsections.

5.1. Impact of facility supply temperature on liquid cooling system performance

In this analysis, outdoor wet-bulb temperature is used as the representative variable for outdoor conditions, as it strongly influences the performance of the cooling towers [28].

As illustrated in Fig. 14, the pump power consumption in the FWS and TCS remains constant, as their flow rates are fixed. In contrast, the power consumption in the CTS, which includes both CTS pumps and cooling tower fans, varies with the outdoor wet-bulb temperature. The CTS pump power remains constant whenever the system is active,

Table 2
Simulated cooling loads and CPU case temperatures under different facility supply temperature setpoints.

Facility supply temperature (°C)	17	22	27
Cooling load (kW)			
From IT equipment	2759.6	2818.6	2877.7
From UPS units	124.6	125.6	126.8
Total cooling load	2884.2	2944.2	3004.5
To liquid cooling system	2806.5	2807.8	2809.3
To air cooling system	77.7	136.4	195.2
Air cooling share (% of IT power)	2.8 %	4.8 %	6.8 %
CPU case temperature (°C)			
Average case temperature	34.4	39.8	45.2
Maximum case temperature	39.6	45.0	50.4

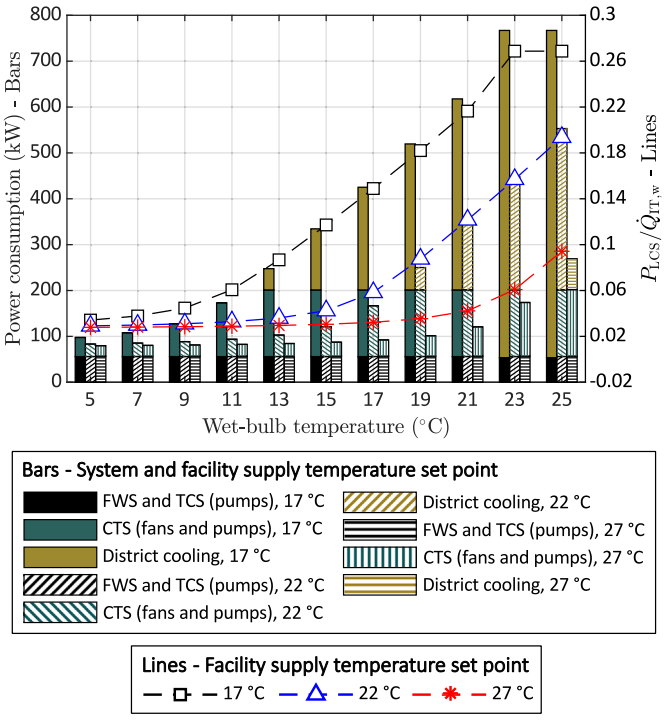


Fig. 14. Power consumption of different components within the liquid cooling system and corresponding inverse energy efficiency ratio ($P_{LCS}/\dot{Q}_{IT,w}$) under varying outdoor wet-bulb temperatures. Simulations are conducted for facility supply temperature setpoints of 17, 22, and 27 °C.

since the flow rate is maintained at a fixed setpoint. Thus, the observed increase in CTS power consumption is solely due to increased fan power.

As the outdoor wet-bulb temperature rises, cooling tower fans must operate at higher speeds to dissipate the cooling load, resulting in increased fan power consumption. Eventually, fans reach their speed limit, indicating that the cooling towers have reached their cooling capacity limit. Beyond this limit, the district cooling system must be activated to handle the remaining cooling load. Following this, the power consumption of liquid cooling system increases more rapidly as the district cooling system is less efficient than the cooling towers. With further increasing wet-bulb temperature, the cooling towers eventually become very inefficient due to the small difference between the facility supply temperature set point and the wet-bulb temperature. In such cases, the CTS is completely deactivated and the cooling load is managed exclusively by district cooling.

Fig. 14 shows that operating at a higher facility supply temperature consistently leads to lower power consumption and higher system efficiency. At lower outdoor temperatures, this advantage is relatively moderate. For example, at a wet-bulb temperature of 5 °C, the liquid cooling system operating at a supply temperature setpoint of 27 °C consumes approximately 79.5 kW, about 17.1 kW lower than at a setpoint of 17 °C. At this lower wet-bulb temperature, all setpoints achieve high efficiency, with IEERs around 0.03, indicating that the power required for heat dissipation is approximately 3 % of the cooling load.

However, the efficiency advantage of a higher setpoint becomes significantly more pronounced at higher temperatures. For instance, at a wet-bulb temperature of 21 °C, the liquid cooling system with a facility supply temperature of 27 °C consumes about 118.6 kW, which is approximately 493.7 kW lower than the system set at 17 °C. This substantial difference arises because, at a 17 °C setpoint, the district cooling system must be activated already at a wet-bulb temperature of around 13 °C. As the wet-bulb temperature reaches 21 °C, district cooling covers 61.3 % of the cooling load at the 17 °C setpoint, whereas at 27 °C, cooling remains fully provided by the cooling towers (100 % free cooling). When the outdoor wet-bulb temperature further increases to 23 °C, this difference grows even larger (to 532.6 kW), as the system with a 17 °C setpoint must entirely deactivate the CTS and rely completely on district cooling, while the 27 °C setpoint still operates using only free cooling.

At a wet-bulb temperature of 25 °C, the IEER for the 27 °C facility supply temperature remains below 0.1, whereas the ratios for the 22 °C and 17 °C setpoints reach approximately 0.2 and 0.27, respectively. This translates into approximately 100 % higher power consumption at the 22 °C setpoint and 170 % higher at the 17 °C setpoint compared to the 27 °C setpoint.

5.2. Impact of facility supply temperature on air cooling system performance

In this analysis, the outdoor dry-bulb temperature is chosen as the primary reference parameter for evaluating air cooling system performance, as it significantly influences the heat dissipation rate of the dry coolers.

Fig. 15 clearly illustrates that the air cooling system consistently operates with lower efficiency than the liquid cooling system, irrespective of outdoor temperatures. This outcome aligns with expectations due to the inherently lower specific heat capacity and density of air compared to water. For example, at a facility supply temperature setpoint of 27 °C and an outdoor wet-bulb temperature of 5 °C, the IEER of the liquid cooling system is approximately 0.03. By contrast, the air cooling system exhibits an IEER of 0.09 (assuming outdoor air is saturated), representing roughly 200 % higher power consumption for the same cooling load.

The efficiency difference between two cooling systems widens further at higher outdoor temperatures. For instance, at a wet-bulb temperature of 25 °C, the IEER of the liquid cooling system is approximately 0.09, whereas that of the air cooling system is 0.33 (assuming saturated air conditions), representing about 267 % greater

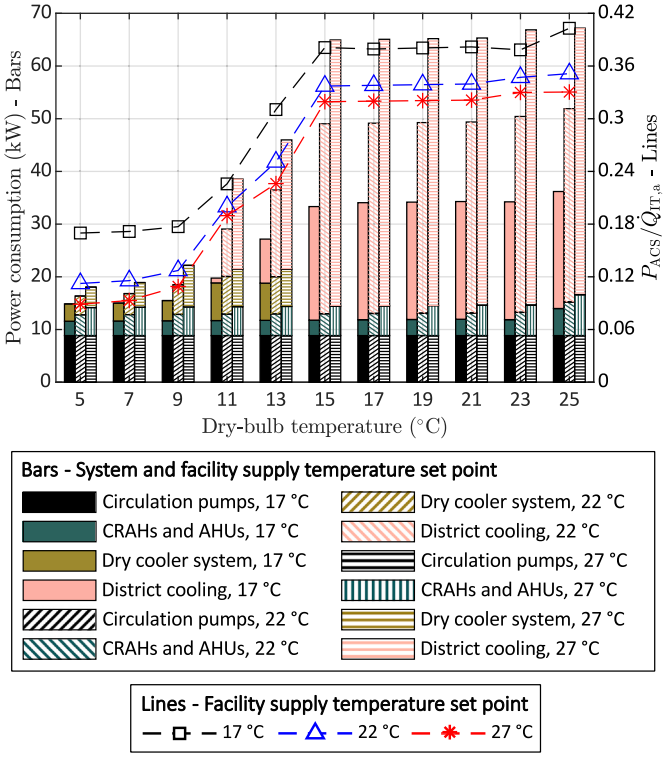


Fig. 15. Power consumption of different components within the air cooling system and corresponding inverse energy efficiency ratio ($P_{ACS}/\dot{Q}_{IT,a}$) under varying outdoor dry-bulb temperatures. Simulations are conducted for facility supply temperature setpoints of 17, 22, and 27 °C.

power consumption. This substantial difference is primarily attributable to the relatively low supply water temperature setpoint of 14 °C required by the CRAH units. Given that dry coolers can cool water only to temperatures approaching the outdoor dry-bulb temperature, which is always higher than the wet-bulb temperature, they reach their operational cooling capacity limit at an outdoor dry-bulb temperature as low as approximately 11 °C. Consequently, starting from approximately 15 °C, the dry cooler system becomes too inefficient and is deactivated, causing the air cooling system to rely completely on district cooling.

Furthermore, the power consumption of the CRAH units increases with higher facility supply temperatures, driven by elevated fan speeds required to handle the increased cooling load on the air cooling system. The power consumption of the AHUs rises at higher outdoor temperatures, primarily due to greater dehumidification demands.

As the cooling load handled by the air cooling system increases with higher facility supply temperatures, its overall power consumption rises accordingly. This difference in power consumption becomes more pronounced at higher outdoor temperatures. For instance, at an outdoor dry-bulb temperature of 5 °C, the air cooling system consumes approximately 17.3 kW at a facility supply temperature setpoint of 27 °C, which is about 3.0 kW (21.0 %) higher than at a setpoint of 17 °C. At an outdoor dry-bulb temperature of 25 °C, the power consumption of the air cooling system increases significantly to around 64.8 kW at the 27 °C setpoint, which is about 31.3 kW (93.4 %) higher compared to operation at 17 °C.

5.3. Impact of facility supply temperature on overall system performance

Fig. 16 illustrates the total power consumption of the HAWK supercomputer system under varying outdoor temperature conditions for the three setpoints of facility supply temperature (17, 22, and 27 °C). By integrating the power consumption of all subsystems, it becomes

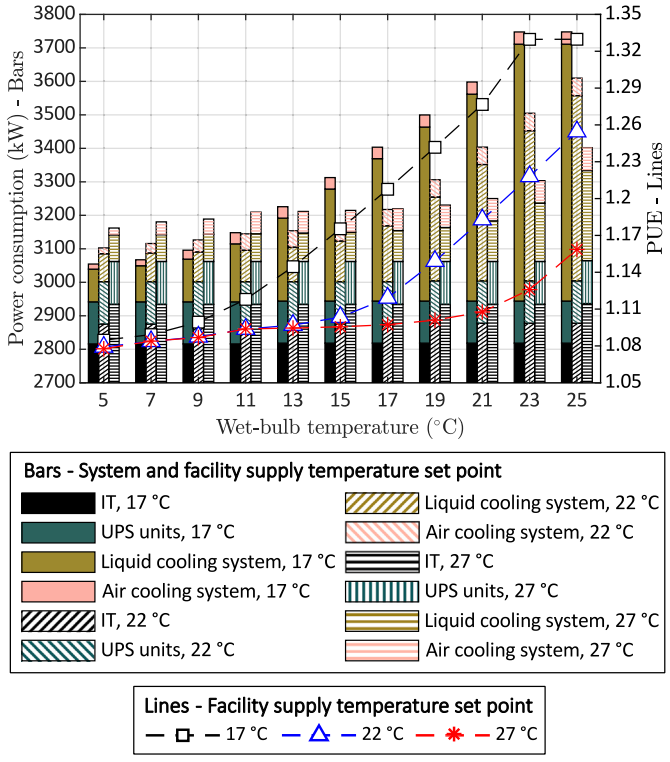


Fig. 16. Power consumption of the individual systems within the HAWK supercomputer and corresponding power usage effectiveness (PUE) values under varying outdoor wet-bulb temperatures (relative humidity fixed at 60 %). Simulations are conducted for facility supply temperature setpoints of 17, 22, and 27 °C. PUE is calculated as the ratio of the total power consumed by the data center to the power consumed by the IT equipment [33].

evident that, for each outdoor temperature, there exists an optimal facility supply temperature that minimizes total power consumption. Furthermore, this optimal setpoint increases with rising outdoor temperature.

For example, at outdoor wet-bulb temperatures below approximately 11 °C, the lowest facility supply temperature setpoint (17 °C) results in the lowest total power consumption. Specifically, at a wet-bulb temperature of 5 °C, the total power consumption at the 17 °C setpoint is approximately 2995.4 kW, which is 108.1 kW (3.5 %) lower compared to the 27 °C setpoint (3103.5 kW).

For outdoor wet-bulb temperature ranges between approximately 13 to 17 °C, the facility supply temperature setpoint of 22 °C achieves the lowest overall power consumption. As the outdoor wet-bulb temperature continues to rise above this range, increasing the supply temperature setpoint to 27 °C becomes more energy-efficient and yields the lowest total power consumption. For instance, at a wet-bulb temperature of 23 °C, the system operating at a 27 °C setpoint consumes approximately 3336.7 kW of power, representing a reduction of 338.8 kW (9.22 %) compared to the 17 °C setpoint (3675.5 kW).

These findings demonstrate that implementing a dynamic control strategy that adjusts the facility supply temperature in response to varying outdoor conditions offers substantial potential for reducing overall energy consumption. Achieving such optimization, however, requires a holistic analysis that considers the interactions between all essential systems within the data center.

Another noteworthy observation is that higher facility supply temperature setpoints consistently result in lower PUE values. However, as demonstrated, a higher facility supply temperature setpoint does not always correspond to lower total power consumption. Therefore, it is crucial to interpret PUE carefully, as lower PUE values alone may not accurately reflect improved total system energy efficiency.

In addition, a simple calculation was performed for the scenario of implementing 100 % heat reuse from the liquid cooling system. Under this condition, the cooling tower system could be fully deactivated, thereby eliminating the power consumption of its fans and pumps. The results show that at an outdoor wet-bulb temperature of 5 °C, the total data center power consumption for facility supply temperatures of 17, 22, and 27 °C would be 2954.7, 3017.2, and 3080.6 kW, respectively. This corresponds to an increase in total power consumption of 2.12 % and 4.26 % when the supply temperature is increased by 5 and 10 K from 17 °C, respectively. It is important to note that, as shown in Table 2, the total heat dissipated to the liquid cooling system remains nearly constant across the three supply temperature setpoints. This raises a critical question: does the energy saved on the heat reuse side offset the increased energy consumption in the data center? Addressing this requires an integrated analysis encompassing both data center operation and heat reuse infrastructure.

6. Discussion

6.1. Generalizability of the findings

The observed trade-off between the power consumption of the liquid cooling system, air cooling system, and IT equipment with increasing facility supply temperature is generally applicable to liquid-cooled data centers with controlled server room environments. This trade-off arises from fundamental physical principles that are not specific to the system studied. The thermal and power modeling approach presented in Section 4 is broadly applicable to other direct liquid-cooled data centers, as it captures the key dynamics as well as the essential cross-system interactions that govern overall data center performance. Furthermore, the implications discussed in Sections 6.2 to 6.4 are relevant to other liquid-cooled data centers, including both direct liquid cooling and immersion cooling. These insights offer guidance for data center power and thermal modeling, liquid cooling system control and operations, and integration with heat reuse.

However, generalizing the quantitative results presented in Section 5 to other facilities requires careful consideration of site-specific factors. For instance, the impact of elevated supply temperature on liquid cooling system efficiency strongly depends on the topology and design of the heat rejection system (e.g., direct-contact cooling towers, dry coolers, or vapor-compression chillers) as well as its design parameters such as nominal cooling capacity, approach temperature, and fan configuration. The relationship between supply temperature and processor case temperature is dependent on the chip-level cooling technology (i.e., whether a single-phase or two-phase liquid cooling approach is employed). The temperature dependence of IT power consumption varies with hardware characteristics, including processor type and memory configuration. Furthermore, the magnitude of the increase in heat dissipated into the server room air depends on rack and enclosure design (e.g., insulation and containment). Consequently, although similar trends and system interactions are expected across other liquid-cooled data centers, the magnitude of the combined effects remains highly site-specific.

6.2. Implications for data center power and thermal modeling

Accurate modeling is essential for effective optimization of data center operation. Based on the analyses conducted in this study, the following key insights regarding modeling process have been identified:

(1) Holistic, cross-system modeling is recommended and can be particularly important in cases where the supply water temperature strongly influences heat transfer to the server room air and IT power consumption. Operational data from the HAWK supercomputer (Section 3) demonstrate that increasing the facility supply temperature improves the energy efficiency of the liquid cooling system. For example, at an outdoor wet-bulb temperature of 19 °C, the dynamically controlled temperature setting (approximately 25 °C) reduced liquid cooling system power consumption by 63.3 % compared to the fixed setting (17 °C). However, this came with an increased heat transfer to the server room air, nearly doubling the proportion of heat dissipated to air from approximately 2.6 % to 5.0 %. Additionally, IT power consumption increased by about 3.15 % when the facility supply temperature was increased from 17 to 25 °C, corresponding to approximately a 3.94 % increase per 10 K. These findings clearly demonstrate that the operation of the liquid cooling system has a considerable impact on both IT power consumption and heat dissipation to the server room air. Although the magnitude of these effects is site-specific, modeling approaches that capture these interdependent dynamics are important to enable reliable data center–level optimization.
(2) Incorporating measurement data into the modeling process is essential for model accuracy and operational safety. As shown in Section 4.3, the calculated cooling capacity of the cooling towers and the overall heat transfer coefficient of heat exchangers under nominal conditions were considerably lower than the values specified in product catalogs. These deviations can result from effects such as fouling, aging, or off-design operation. Relying exclusively on product specifications may lead to overestimating cooling performance, thus increasing risks of overheating and operational failures, especially in model-based control scenarios.

6.3. Implications for control and operation of the liquid cooling systems

Improving the control and operation of the liquid cooling system is vital to fully realize its energy-saving potential. The following insights are particularly relevant for developing effective control strategies and operational practices for liquid-cooled data centers:

(1) Liquid cooling system can act as a key lever for data center-level energy optimization. Simulation results in Section 5.3 demonstrate that at a wet-bulb temperature of 5 °C, operating the HAWK supercomputer at a 17 °C facility supply temperature resulted in a total power consumption of 2995.4 kW, 108.1 kW lower than at a 27 °C setpoint. Notably, the liquid cooling system itself consumed only 79.5 kW at the 27 °C setting, indicating that its influence on total power extends beyond its own consumption due to interactions with other systems. This finding suggests that optimized control of the liquid cooling system can offer substantial potential for reducing overall data center energy consumption.
(2) The simulation analysis in Section 5.3 also shows that implementing a dynamic control strategy that adjusts the facility supply temperature in response to varying outdoor conditions offers considerable potential for energy savings. In other words, for a given outdoor condition, there exists an optimal facility supply temperature that minimizes total power consumption. Additionally, this optimal setpoint increases with rising outdoor temperature.
(3) PUE is not suitable as the sole control objective. PUE is widely adopted as a standard efficiency metric in data center operations, nevertheless, it may not accurately reflect actual energy performance. As shown in Fig. 16, although higher facility supply temperatures consistently yielded lower PUE values, a lower PUE did not always indicate a lower total energy consumption. Therefore, careful interpretation of PUE values is necessary, as they alone may not reliably indicate improved overall energy efficiency. These findings are consistent with observations reported in studies on air-cooled data centers [34,35].

6.4. Implications for integration with waste heat reuse applications

Data centers are increasingly being investigated for waste heat reuse. It is well known that raising the temperature level in a data center's cooling system improves the energy efficiency of the heat reuse infrastructure, thereby enhancing the feasibility of waste heat reuse applications. However, elevated cooling system temperatures can also lead to increased power consumption within the data center itself, potentially offsetting the gains achieved on the heat reuse side. As discussed in Section 5.3, in the case of the HAWK supercomputer, assuming 100 % heat reuse from the liquid cooling system, increasing the facility supply temperature from 17 to 27 °C led to a 4.26 % rise in total data center power consumption at an outdoor wet-bulb temperature of 5 °C. Notably, the total heat transferred to the liquid cooling system remained nearly constant, despite the rise in IT power caused by the higher supply temperature. The critical question that arises here is whether the higher temperature in the liquid cooling system yields greater energy savings on the heat reuse side than the additional energy consumed within the data center. Furthermore, under given outdoor and operational conditions, what facility supply temperature setpoint achieves an overall energy efficiency optimum when both the data center and heat reuse infrastructure are considered? Addressing this question requires an integrated modeling and analysis approach that encompasses both the data center and the heat reuse infrastructure.

6.5. Limitations and future work

Despite the comprehensive scope of the developed model, several limitations remain. First, power losses from power distribution units were included in IT power, as explicit measurements were not available. Explicit modeling of these losses would be preferable, nevertheless, their inclusion ensures plausible total power consumption results. Second, since direct measurements of CPU case temperature and utilization were not available, assumptions were necessary during the model fitting process, introducing additional uncertainty into the parameterization of the IT power model. Third, some models were derived using specifications from product catalogs and planning documents, which may lead to deviations in predicted energy performance.

Nevertheless, the model effectively captures the essential dynamics and complexity of thermal behavior and energy performance in direct liquid-cooled data centers. The developed model provides a foundation

for future research on advanced cooling control strategies to optimize energy efficiency and on heat reuse in direct liquid-cooled data centers.

7. Conclusion

Further improvements in the energy efficiency of direct liquid-cooled data centers require advanced cooling control strategies capable of dynamically responding to varying operational and environmental conditions. A key parameter for optimization within the liquid cooling system is the supply water temperature. This study presents a comprehensive analysis of the impact of supply water temperature in the liquid cooling system on the energy performance of a direct liquid-cooled data center, based on extensive measurement data and holistic modeling.

First, operational data from the direct liquid-cooled HAWK supercomputer at the High-Performance Computing Center Stuttgart, collected over a 21-month period, were analyzed. The analysis revealed that increasing the supply water temperature improves the energy efficiency of the liquid cooling system. However, this improvement is accompanied by increased IT power consumption and greater heat transfer to the server room air. These interactions highlight the importance of cross-system analysis.

Based on insights gained from operational data analysis, a data center-level power and thermal model was developed for the HAWK supercomputer. This holistic model integrates key dynamics, including the thermal behavior and power consumption of the liquid and air cooling systems, the temperature dependency of IT leakage power, and the heat transfer from IT equipment to the server room. Due to the availability of extensive measurement data, the essential dynamics of most system components could be accurately identified and modeled. For components without direct measurements, models were developed using detailed information from product catalogs, planning, and commissioning documentation. Validation against one year of hourly measurement data demonstrated high prediction accuracy, with a CVRMSE of 11.52 % for the liquid cooling system's power consumption. Moreover, the presented thermal and power modeling approach is broadly applicable to other direct liquid-cooled data centers, as it captures the key dynamics as well as the essential cross-system interactions that govern overall data center performance.

Using the developed model, simulation studies were conducted to analyze the impact of different supply water temperature setpoints on the energy performance of individual subsystems and the overall data center. The key insights derived for liquid-cooled data centers in general are summarized as follows: (1) The liquid cooling system can act as a key lever for data center-level energy optimization, influencing total power consumption beyond its own direct power consumption due to its interactions with the IT and air cooling systems. (2) Dynamically adjusting the supply water temperature in response to outdoor conditions offers considerable potential for energy savings. (3) PUE does not reliably reflect overall energy performance and should not be used as the sole control objective. (4) Elevated supply temperatures can enhance the efficiency and feasibility of waste heat reuse but may also increase the data center's power consumption. Optimal operation requires integrated analysis that accounts for both data center and heat reuse infrastructure.

In summary, this work provides practical guidance for improving the operation of liquid-cooled data centers and highlights the importance of holistic modeling in understanding cross-system interactions. The developed model effectively captures the essential dynamics and complexity of thermal behavior and energy performance in direct liquid-cooled data centers and offers a robust foundation for future research on advanced cooling control strategies and heat reuse integration.

CRediT authorship contribution statement

Mashhur Gheni: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Henner Kerskes:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Konstantinos Stergiaropoulos:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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